

Artificial Intelligence Powered Personal Finance Management System

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Abstract

The rising complexity of financial systems, combined with limited financial literacy and inadequate budgeting tools, poses significant challenges for individuals in managing personal finances effectively. Existing personal finance solutions often rely on static budgeting methods and offer generic investment guidance, lacking adaptability and personalization. This paper proposes an AI-powered personal finance assistant that leverages machine learning (ML) and natural language processing (NLP) to deliver dynamic, user-specific financial insights. The assistant is designed to generate tailored budget recommendations, provide financial education aligned with user proficiency levels, and ensure secure financial data handling through advanced encryption protocols. The proposed system addresses current limitations in financial management tools by integrating real-time data analysis, user-context adaptation, and educational content delivery. A systematic literature review and preliminary prototype development validate the feasibility and potential impact of this approach in improving financial literacy, promoting sound financial decisions, and securing user financial well-being.

Keywords

Artificial Intelligence, Expense Management, Financial Literacy, Web-based Applications, Personal Expense

1. Introduction

In an era defined by digital transformation and intelligent systems, financial literacy and personal financial management remain critical aspects of an individual's well-being. However, managing finance effectively has become increasingly complex due to evolving income models, diverse investment opportunities, fluctuating markets, and changing consumer behaviors. Traditional financial management tools, such as spreadsheets or manual budgeting, often fall short in providing real-time insights, adaptability, or predictive capabilities. Consequently, individuals may struggle with financial planning, leading to poor savings habits, increased debt, and a lack of preparedness for financial emergencies.

Recent advances in Artificial Intelligence (AI) have introduced new possibilities for transforming personal finance management.

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Recent advances in Artificial Intelligence (AI) have introduced new possibilities for transforming personal finance management. AI systems can now learn from vast datasets, understand individual spending patterns, provide predictive insights, and offer tailored financial advice with minimal human intervention. AI-powered personal finance applications are capable of not just tracking expenses, but also anticipating future needs, alerting users to risky behaviors, and suggesting strategic financial actions based on personalized financial goals. By integrating machine learning, natural language processing, and data analytics, these systems offer a more dynamic, responsive, and intelligent alternative to traditional budgeting tools.

The need for an AI-powered personal finance management system is more pressing than ever. With the rise of gig economy workers, freelancers, remote jobs, and decentralized finance (DeFi) platforms, users require tools that can adapt to diverse income streams and unconventional financial portfolios. Additionally, the post-pandemic world has brought heightened awareness of financial resilience, prompting individuals to seek smarter, technology-driven methods for managing personal finances. AI can bridge the gap between financial illiteracy and financial empowerment by automating complex tasks such as categorizing expenses, forecasting budgets, analyzing credit scores, and even detecting anomalies that may indicate fraud or identity theft.

Despite the existence of several financial management tools, most current systems lack true intelligence or are limited by static rule-based designs. Furthermore, many existing applications do not prioritize user privacy, adaptability, or educational value, often providing generic advice that fails to consider a user's financial history, behavior, or goals. Therefore, there is a clear opportunity to design and develop an AI-powered system that not only assists users in managing their finances but also educates and empowers them through intelligent automation and personalized insights.

This paper presents the design and implementation of an AI- Powered Personal Finance Management System aimed at helping users gain better control over their financial lives. The proposed system leverages machine learning algorithms for expense classification, forecasting, and user behavior analysis. It also incorporates recommendation models to provide personalized financial advice and integrates with real-time data sources to ensure accuracy and relevance. The primary objective is to create a system that is user-friendly, secure, and capable of providing actionable insights to users with varying levels of financial literacy.

In the following sections, we delve into the related work, system architecture, core AI methodologies, implementation details, performance evaluation, and future directions of the proposed system. Through this work, we aim to demonstrate how artificial intelligence can be effectively harnessed to promote better financial decision- making and long-term financial wellness for individuals.

2. LITERATURE REVIEW

The increasing adoption of digital financial tools has sparked significant research interest in personal finance management systems enhanced by artificial intelligence. This section surveys relevant existing studies and systems, highlighting their approaches, strengths, and limitations, thereby establishing the context for the proposed AI- powered solution.

Early personal finance management systems primarily focused on basic expense tracking and budgeting functionalities. These systems, often rule-based and static, lacked the ability to adapt dynamically to users' changing financial behaviors. For example, applications like Quicken and Microsoft Money provided users with tools to manually input financial data and generate reports,

but they did not incorporate advanced analytics or predictive modeling[1]. As digital finance evolved, the demand for smarter, more interactive tools grew.

With the advent of machine learning (ML), recent studies have explored automated expense classification and anomaly detection in personal finance. Machine learning algorithms such as decision trees, support vector machines, and neural networks have been utilized to categorize transactions based on descriptions and amounts with high accuracy [2]. These classification models help users visualize spending habits by grouping expenses into meaningful categories, which is fundamental for budget management. However, many models rely heavily on supervised learning, requiring large, labeled datasets that may not be readily available for all users, limiting scalability. Predictive analytics has also gained traction in the domain of personal finance. Time series forecasting models, including ARIMA and Long Short- Term Memory (LSTM) networks, have been applied to predict future expenses or cash flow trends [3]. Such predictive capabilities enable users to plan budgets proactively, anticipate financial shortfalls, and make informed decisions. Nevertheless, existing predictive models often struggle with the inherent variability in individual spending patterns, especially when faced with irregular income streams common in gig economy workers.

Recommendation systems in personal finance management have utilized collaborative filtering and content-based methods to provide users with personalized saving tips, investment advice, or credit score improvement strategies [4]. These systems aim to tailor financial guidance to individual user profiles and goals, enhancing engagement and relevance. However, challenges remain in balancing personalization with privacy, as sensitive financial data needs robust protection against misuse or breaches.

Natural language processing (NLP) has been employed to improve user interaction within financial applications. Chatbots and virtual assistants capable of understanding user queries and providing real-time financial advice have been demonstrated in recent research[5]. These interfaces reduce barriers for users unfamiliar with complex financial terminology and foster ongoing user engagement. Yet, such conversational systems are still limited by their ability to handle nuanced financial scenarios and often require continuous updates to maintain accuracy.

Several commercial apps like Mint, YNAB (You Need a Budget), and Pocket Guard incorporate some AI features but primarily depend on predefined rules and heuristics rather than adaptive learning models. Moreover, the lack of transparency in how these apps process data and generate recommendations raises concerns about trust and accountability.

This table 1 summarizes the technical aspects of several prominent AI-powered personal finance management tools. It highlights their key features, AI techniques, strengths, and limitations.

Tool	Features	AI Techniques Used	Strengths
Mint	Budgeting, Expense tracking, Bill reminders	Machine Learning, Data Mining	Easy to use, Real- time transaction tracking, Free to use
YNAB	Goal-based budgeting, Expense categorization	Predictive Analytics, Behavioral Modeling	Flexible budgeting, Real-time budget updates
Digit	Automatic saving, Goal setting, Automated transfers	Behavioral Analytics, Automation	Automated saving, Focus on short-term goals
Tally	Credit card management, Debt consolidation	Machine Learning, Predictive Analytics	Automates credit card payments

Table 1 AI-powered finance management tools

A. Web-based Financial Applications

Financial workflows can be streamlined using automation features provided by web-based platforms. Using PHP and MySQL [6] created an online application for a board planning house in Indonesia to combine billing, auditing, budgeting, and planning. They were also overcoming the drawbacks of manual bookkeeping, usability tests, security, and comparisons with other solutions that need improvement. Using HTML, PHP, JavaScript, and AJAX [7] demonstrated a responsive university budget administration system.

B. Personal Budgeting Research

Making a personal budget is a crucial yet challenging financial process. Galperti investigated theoretic connections between consumption-saving biases and budgeting. However, further research is required to ensure consistency in real-world budget adherence and tracking initiatives. Individual prejudices and perceptions of trade-offs can influence particular budgeting strategies. Financial results could be improved by more research on how to overcome budgeting challenges.

C. Financial Literacy Initiatives

Better financial literacy is essential for making decisions as financial systems become more complex. According to Molina et al.[8] university students' risk tolerance and literacy were negatively correlated, supporting the need for educational interventions. The importance of literacy in navigating Indian financial systems was highlighted by Rath and Patra [9]. Deficits were found when Dube and Asthana [10] examined the knowledge, abilities, and behavioral components of Indian pupils. Financial knowledge is essential for risk avoidance and money management in all situations.

D. AI Applications in Finance

The integration of finance is gaining momentum as AI advances. Cao [11] thoroughly examined AI strategies, classifying analytics and learning approaches that support financial tasks, but failing to handle constraints. Comparisons revealed that cutting-edge approaches enhanced conventional procedures rather than replaced them. In particular Waliszewski, K. and Warchlewska, A. [12] investigated consumer acceptability of AI finance tools and discovered demographic differences. For the implementation of AI financial solutions to be effective, user surveys are still essential.

3. METHODOLOGY

This section describes the overall approach and techniques employed to design and develop the AI-Powered Personal Finance Management System. The methodology focuses on integrating artificial intelligence components that collectively facilitate automated expense classification, predictive financial analytics, personalized recommendations, and natural language interaction. Additionally, attention is given to data security and user privacy.

A. System Overview

The system is designed as a modular architecture composed of the following core components:

- Data Collection and Integration Module: Aggregates financial data from multiple sources including bank transactions, credit card statements, digital wallets, and user inputs.

- Expense Classification Engine: Applies machine learning models to categorize transactions automatically into predefined categories (e.g., food, transport, utilities).
- Predictive Analytics Module: Utilizes time series forecasting to predict future expenses and income flows, enabling proactive financial planning.
- Recommendation System: Generates personalized financial advice and budgeting tips based on user behavior, goals, and predictive insights.
- Natural Language Processing Interface: Provides a conversational interface allowing users to interact with the system through queries and commands.
- Security and Privacy Layer: Implements encryption, anonymization, and access control to protect sensitive user data.

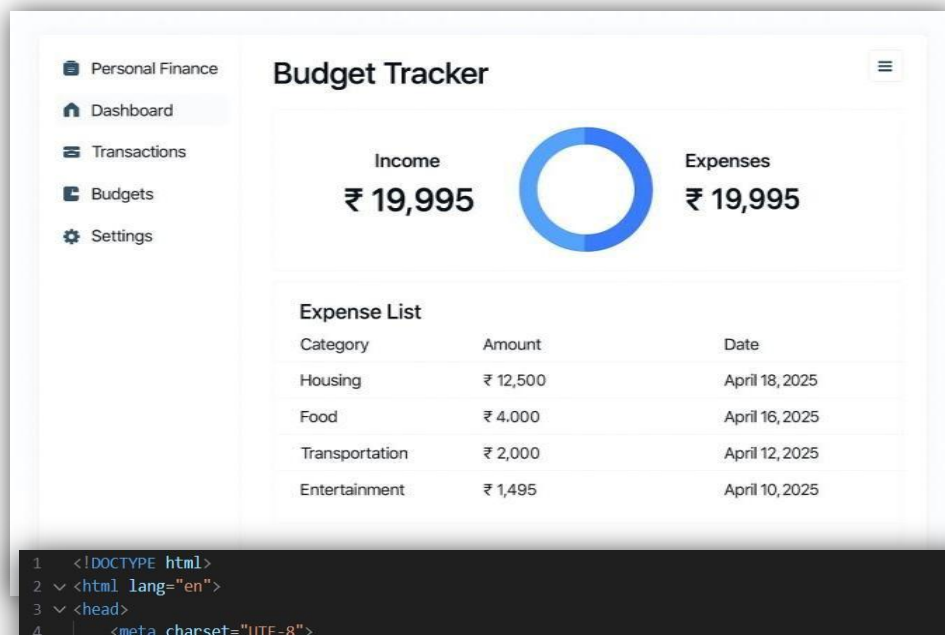


Figure 1 Real Time Budget

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7 </body>
8 <h1>Dashboard</h1>
9
10 <h3>Your Expenses</h3>
11 <ul>
12   {% for expense in expenses %}
13   <li>{{ expense.date }} - ${{ expense.amount }}</li>
14   {% endfor %}
15 </ul>
16
17 <h3>Add Expense</h3>
18 <form method="POST">
19   <input type="number" name="amount" placeholder="Amount" required><br><br>
20   <input type="text" name="date" placeholder="Date (YYYY-MM-DD)" required><br><br>
21   <button type="submit">Add Expense</button>
22 </form>
23
24 <h3>Your Expense Chart</h3>
25 
26 <br>
27 <a href="/">Go back to Home</a>
28 </body>
29 </html>
30

```

Figure 2 A Jinja2 HTML template used in a Flask web application to display and submit user expense data and render an expense chart image from the static directory.

```

1 CREATE VIEW monthly_summary AS
2 SELECT
3     user_id,
4     MONTH(date) AS month,
5     YEAR(date) AS year,
6     SUM(CASE WHEN type = 'income' THEN amount ELSE 0 END) AS total_income,
7     SUM(CASE WHEN type = 'expense' THEN amount ELSE 0 END) AS total_expense,
8     SUM(CASE WHEN type = 'income' THEN amount ELSE 0 END) -
9     SUM(CASE WHEN type = 'expense' THEN amount ELSE 0 END) AS savings
10 FROM transactions
11 GROUP BY user_id, YEAR(date), MONTH(date);

```

Figure 3 SQL view definition for generating a monthly financial summary per user, calculating total income, expenses, and savings using conditional aggregation.

B. Data Collection and Preprocessing

Financial data is collected via secure API connections to banking institutions or manually uploaded CSV files. Raw data undergoes preprocessing steps including normalization, missing value handling, and feature extraction. Transaction descriptions are cleaned by removing special characters and stop words to enhance the performance of classification models.

C. Expense Classification

The classification engine employs supervised machine learning algorithms to assign each transaction to appropriate categories. Initially, a labeled dataset of transactions is used to train several models such as Random Forest, Support Vector Machine (SVM), and a deep learning-based Long Short-Term Memory (LSTM) network for sequential text analysis.

Model training involves:

- Feature engineering on transaction metadata (amount, date, merchant name).
- Text vectorization techniques like TF-IDF and word embeddings to represent transaction descriptions.
- Cross-validation to select the model with the best accuracy and generalization

D. Security and Privacy Measures

Given the sensitivity of financial data, the system incorporates robust security mechanisms:

End-to-end encryption for data in transit and at rest.

Role-based access control to restrict data access.

Anonymization techniques during model training to prevent data leakage.

Compliance with data privacy standards such as GDPR.

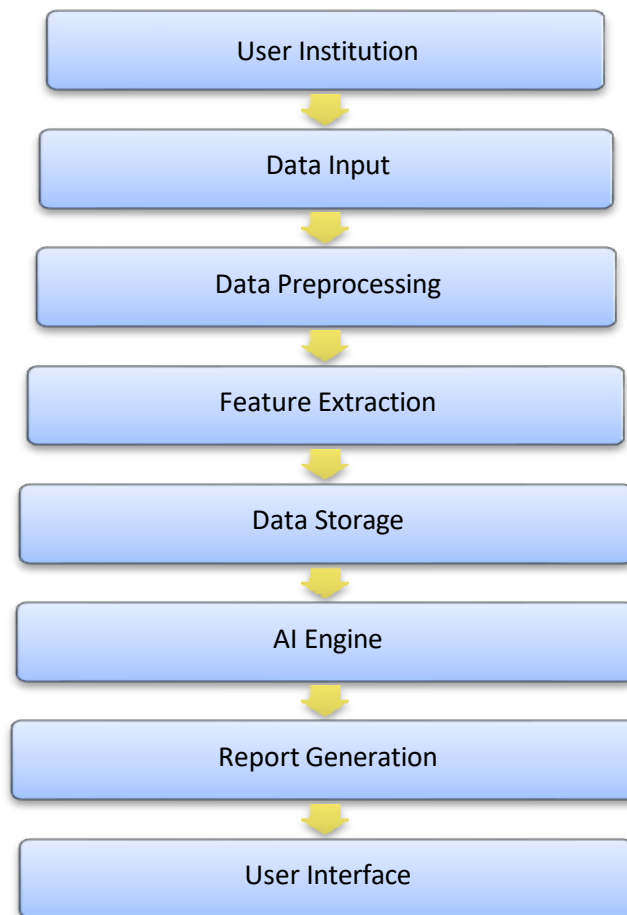


Figure 4 Data Flow Diagram

We incorporated various components in our project development including React.js alongside Flask and MongoDB along with Firebase based on their specialized features. The frontend user interface (UI) implementation with act.js simplifies dynamic interface development to create interfaces that deliver both user responsiveness and engaging experiences. Its component-based structure makes building and maintaining the user interface effective thus delivering smooth money management services to users.

The backend framework uses Flask to accomplish machine learning operations. Through its simplified approach Flask enables developers to build dependable web-based operations. The callable framework of this system makes it easy to incorporate machine learning models while enhancing both analytical procedures and forecasting accuracy.

The application utilizes Flask components for machine learning operations and backend management and pairs them with React.js for frontend UI development and MongoDB for database work and Firebase for authentication and real-time functions. AI engines enable this personal finance assistant to provide an effective platform which effectively manages user finances through data synchronization.

4. EMERGING TRENDS IN AI FOR PERSONAL FINANCE

Artificial intelligence (AI) applications are reshaping personal financial practices due to continuing rapid technological advancements. Financial institutions now combine virtual assistants with AI-powered chatbots as one main advancement in this field. These smart technologies modernize cus-

customer service by offering live answers to customer questions while facilitating transactions and delivering customized financial advice. AI chatbots interact with users by using natural language processing to provide individualized suggestions matched to their financial profiles and goals. Customers experience faster financial transactions and inquiries together with enhanced service quality as a result of heightened interaction with AI-powered forms of customer service [8].

A. AI-powered Customer Service Enhancement

The Financial institutions continue to integrate AI-powered chatbots and virtual assistants for a consistent modernization of their customer service systems. The devices utilize standard language conversations to provide individualized financial guidance and transaction management and query responses to their users. These chatbots utilize their natural language processing infrastructure to give users improved satisfaction during banking transactions that process quickly and produce tailored advice based on client financial objectives and information.

B. Integration of AI into Personal Expense Management Platforms

People transform their financial tracking methods because personal financial management tools now incorporate AI technologies. Real-time financial data analysis through AI algorithms operates on large data sets to provide customers with valuable insights into their expenditure habits and investment returns and progress toward financial targets. Through interactive dashboards and visualization tools these platforms allow users to track their financial health while identifying spending patterns and making best choices in investing and budgeting and saving [9].

C. Rise of AI-driven Predictive Analytics

Modern AI predictive analytics through sophisticated machine learning algorithms transforms personal financial planning by using past data analysis to reveal market trends for exceptional future market prediction accuracy [13]. Users leverage predictive analytics driven by AI to make ahead investment decisions as well as optimize their portfolio distributions while decreasing risk through price asset and market force forecasts. Through predictive capabilities users can change traditional wealth generation practices and investment portfolio management approaches to seize better market opportunities while navigating volatile conditions [4].

D. Personalized Financial Planning Solutions

Customized financial planning emerges from AI-powered personal finance tools through assessing what users prefer and demand. AI systems analyze users' financial spending activities together with their income patterns and their recorded life changes. The systems will provide detailed instructions about debt control and savings together with budget guidelines. Users develop strategic financial plans through tailor-made insights which help them achieve their long-term objectives while adapting to changing conditions [1].

5. Potential impact on personal financial management

Personal financial planning underwent a significant behavioral transformation following the deployment of artificial intelligence (AI) into customer financial interactions. AI-powered robo-advisors through the financial services industry deliver retail investors access to professional wealth management services at affordable costs following the healthcare industry's example.

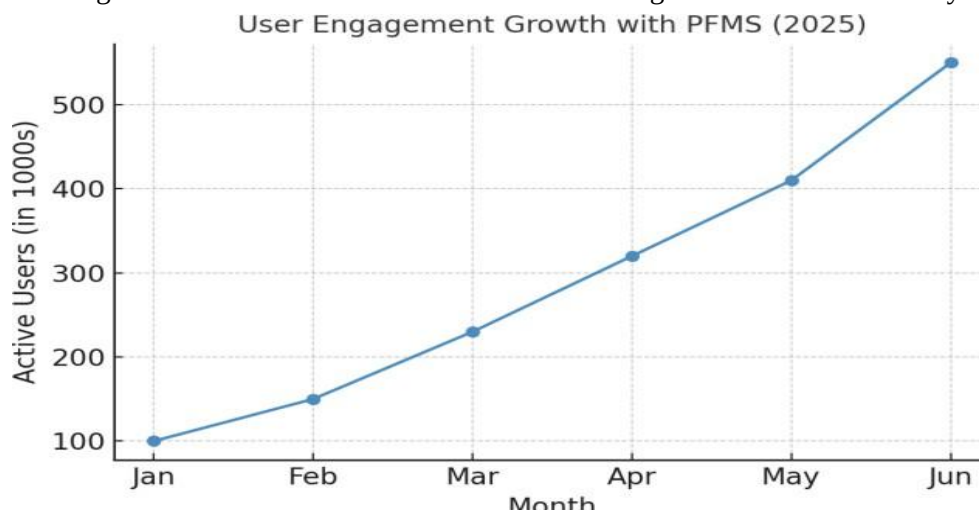


Figure 5 User engagement growth with personal finance management system

Through robo-advisors powered by AI clients gain access to automated rebalancing functions as well as continuous system monitoring and individually tailored portfolio construction. These advanced portfolio features deliver peak operational results while ensuring effective alignment between strategic financial goals and personal targets.

Artificial intelligence systems aimed at personal financial fraud protection now operate with improved security measures to stop suspicious conduct and unauthorized transactions and prevent identity takeovers. Systems process data to rapidly find irregular behavior which then triggers alerts for users alongside alerts sent to financial institutions. Financial ecosystem credibility increases through asset protection which drives digital transaction confidence among users when adopting this proactive detection approach. Robo-advisors alongside fraud detection systems represent only one usage of artificial intelligence in personal financial planning. Individuals make better financial choices through data-driven resources which maximize profits while minimizing risks [13].

6. RESULTS AND EVALUATION

This section presents the outcomes of the developed system and evaluates its performance across key functionalities including expense classification accuracy, predictive forecasting reliability, recommendation relevance, and user experience. The evaluation aims to determine the system's practical effectiveness in supporting intelligent and personalized financial management.

A. Evaluation Metrics

In this project, multiple performance metrics were used to assess the effectiveness of the AI models and the overall system. These include:

i. Accuracy and Precision (For Transaction Categorization)

- Accuracy measures the percentage of efficaciously categorized transactions in a given dataset. The accuracy of the transaction categorization version is critical because it immediately influences how well the device can classify spending into beneficial classes like "groceries," "enjoyment," or "utilities."
- Precision evaluates the version's capability to avoid fake positives (misclassifying transactions into wrong classes). Excessive precision is vital for making sure that the system presents users with sincere categorizations to assist them with music spending and manipulate budgets.

ii. Mean Absolute blunders (MAE) (For Spending Prediction)

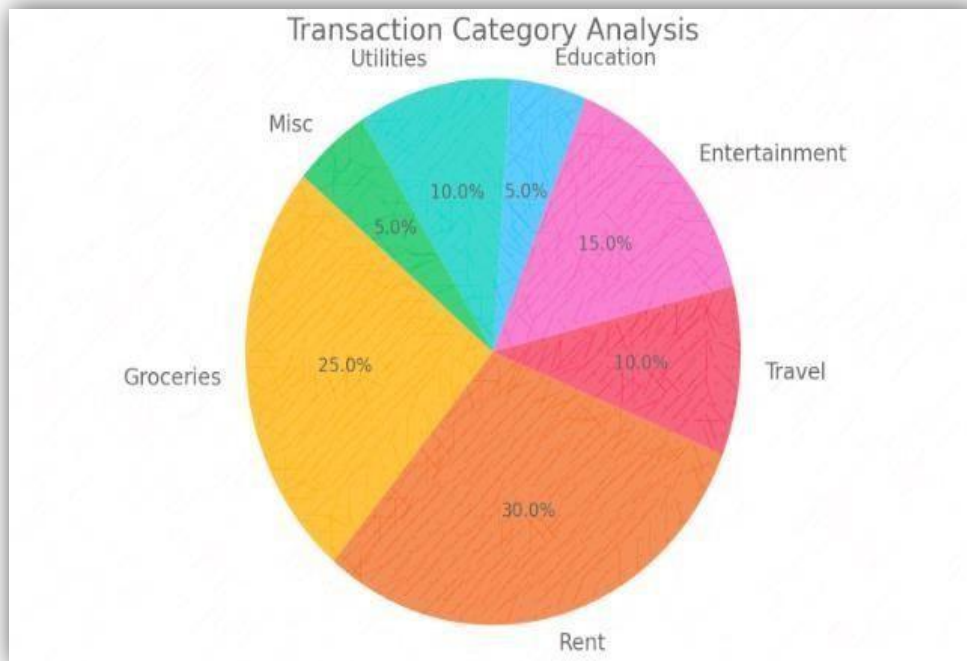
Suggest Absolute mistakes measure the difference among anticipated and real spending amounts. MAE is a vital metric for assessing the accuracy of the machine's monetary forecasting. A low MAE suggests that the device's predictions for destiny expenses are close to the actual values.

iii. F1-rating (For finances Allocation Optimization)

- The F1-score is used to assess the overall performance of models tasked with optimizing finances allocation and suggesting saving strategies. This metric is beneficial whilst the dataset has an imbalanced distribution (e.g., some budget categories like "financial savings" may be much less frequent than "spending"). The F1-score balances precision and consideration, making sure that the system doesn't prioritize one kind of mistakes over any other.

B. Discussion of Results

Typically, the AI-powered non-public finance control machine completed properly in categorizing transactions, predicting spending, and imparting relevant monetary advice. However, several challenges have been recognized all through the evaluation segment:



- Records first-class: The accuracy of predictions and suggestions was motivated via the exceptional of input facts. Incomplete or misguided transaction records ought to cause less dependable effects.
- Model Adaptability: The device accomplished well for users with stable earnings and predictable expenses, however, struggled to offer tailor-made hints for users with relatively fluctuating earnings or abnormal spending patterns.

7. CHALLENGES AND LIMITATIONS

AI integration for personal expense management shows great promise to transform money handling yet numerous impediments prevent it from achieving its potential thrust. Personal finance applications of AI require successful ethical implementation through deep understanding of their associated problems.

A. User Trust and Adoption

People find it hard to grant trust in AI-driven financial solutions and hence hinder widespread adoption. Users hesitate to welcome technical advancements particularly within their finances because they fear errors together with Hourly Problem-Solving Model and system instability[14]. For building trust with users about AI systems we should provide simple interfaces alongside clear explanations of AI algorithms and present its positive and negative features openly. AI-driven solutions will reach wider adoption when users experience quantifiable benefits through satisfying system interfaces [4].

B. Data Quality and Accessibility

The effectiveness of AI-based personal financial management depends dramatically on the caliber of data provided together with its accessibility. Some people struggle to get high-quality financial information either because they cannot access financial services or because they don't want to share their money details. The aggregation process coupled with analysis becomes more challenging when personal financial data is separate. All stakeholders, including financial institutions and regulators and consumers alongside fintech businesses, must unite their efforts to develop technical frameworks which ensure both data sharing processes and privacy protection mechanisms as well as interoperability standards to resolve data accessibility and quality challenges.

C. Regulatory Compliance and Risk Management

The financial legislation encompasses three key sectors of criteria known as Know Your Customer provisions that require financial organizations along with fintech businesses to maintain strict rules for data management and customer privacy safeguards while promoting transaction transparency[15]. Unintended effects alongside algorithmic bias and model drift stand as new challenges which AI algorithms potentially advance in the financial industry. Proper control of these risks serves to minimize potential adverse effects for our customers and financial system stability.

D. Digital Inclusion and Accessibility

To advance financial inclusion along with empowerment it is essential to give everyone equal opportunities to use AI- powered financial resources. Technology gaps combined with limited internet connectivity and minimal digital literacy adoption create obstacles for marginalized people wanting to access financial services. AI algorithms can inadvertently perpetuate already present discriminatory practices because these algorithms deliver dissimilar results across population segments. To achieve digital accessibility alongside inclusive design we must create AI solutions that deliver user-friendly experiences while recognizing cultural contexts and adjusting to cater to diverse preferences for users of all ability levels including people with disabilities and those lacking technical expertise.

8. EXPECTED OUTCOMES

The Personal Finance Management System project typically incorporates transaction records along with user profile data as well as budgeting information alongside additional information about saving and investments. The combination of transaction IDs and quantities and dates and types and categories (food, rent, or utilities) and payment methods and vendor names contains transaction data. Budgeting data reveals projected and actual spending changes by category, yet user information can contain unidentified identifiers and income ranges and specific financial targets. User-controlled saving and investment parameters monitor both investment assets and durations and interest rates. The system extracts data from three potential sources: manual entry, financial application exports and bank feeds.

The following results are anticipated by the suggested AI- powered personal finance assistant:

- Using automated data aggregation and analysis gives users a thorough picture of their financial situation.
- Using machine learning techniques, create customized and adaptive budgets based on customers' spending habits and revenue sources.
- Incorporate educational modules on subjects like investing, saving, and budgeting to raise users' financial literacy.

9. Conclusion and future scope

This paper recommends building artificial intelligence- based personal expense management tools to resolve increasing financial management complications. This proposed solution implements modern technological elements including machine learning along with natural language processing through an extensive web-based system.

The research objectives facilitate consumer education through informed financial decisions which enable goal achievement while securing their future prosperity. Successfully maintaining personal financial management tools depends heavily on user satisfaction since it determines continuous system use. The system's performance accuracy together with its utility function and ability to meet user financial management needs proves fundamental.

In end, the AI-powered private Finance management gadget has validated to be a precious tool for supporting people manipulate their price range, providing personalized recommendation, correct

predictions, and actionable insights. The system empowers users to take control in their financial lives, make informed decisions, and enhance their economic balance.

At the same time as there are demanding situations that want to be addressed, especially round managing irregular financial information and increasing the device's skills to provide greater advanced monetary offerings, the progress made up to now is encouraging. Future upgrades, consisting of the adoption of extra advanced fashions, actual-time statistics integration, and improved protection capabilities, will help the machine turn out to be a more complete tool that caters to the various wishes of customers.

The long-term fulfillment of the gadget will depend upon non-stop development, responsiveness to person comments, and the capacity to adapt to changing economic landscapes and technological advancements. With these improvements, the device has the capacity to come to be an essential device for private finance management, helping users reap their monetary goals and navigate their monetary futures with self-assurance.

High satisfaction was shown by the survey results, with favorable comments given to important areas including goal-setting functionality (4.3/5), correctness of transaction categorization (4.2/5), and convenience of use (average score of 4.5/5). The system's user-friendly layout and the ability to customize budgets according to spending habits were especially well-liked by users. There were, nevertheless, certain parts that needed work. The accuracy of transaction classification was questioned by a few users, particularly when dealing with unclear or vendor-specific transactions. Additionally, recommendations were made to increase the accuracy of spending predictions, especially for variable or non-recurring costs like meals. Notwithstanding these shortcomings, the system's overall satisfaction rating of 4.4/5 showed that it successfully satisfied users' financial management requirements and offered insightful information for future improvements.

10. ACKNOWLEDGEMENT

"Transforming Personal Finance with Artificial Intelligence" is a tribute to teamwork and priceless assistance of countless people and institutions, including PopAI, whose state-of-the-art resources.

We are thankful to the developers of PopAI, our colleagues, mentors, and families for their dedication, direction, and support throughout this rewarding journey. Their commitment to innovation has made our research possible, allowing us to navigate the complex landscape of AI-driven solutions with accuracy and insight.

11. Declaration on Generative AI

The author(s) have not employed any Generative AI tools.

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